

An Integrated Core-and-Cluster Framework for Teaching Physical Examination Aligned with Clinical Reasoning

Background

Physical examination teaching in early medical curricula commonly emphasizes a comprehensive head-to-toe approach. While foundational, it is time-intensive, inconsistently reflected in routine practice, and misaligned with how clinicians select maneuvers in response to specific patient problems. Learners struggle to distinguish elements performed universally from those applied selectively, limiting integration of examination with clinical reasoning.

Summary of Work

We implemented a Core + Cluster physical examination framework, adapted from the model described by Gowda and colleagues, within an early clinical skills curriculum. Faculty synthesized national consensus literature and institutional priorities to define a Core examination of 24 maneuvers, performed for all patients. Three universal items (hand hygiene before and after, and draping) are performed every encounter and, as in published cores, counted separately. We also specified eight domain-based clusters (cardiac, pulmonary, gastrointestinal, neuro-vestibular, musculoskeletal, HEENT, genitourinary, endocrine/systemic), each adding maneuvers selected to support hypothesis-driven clinical reasoning. All maneuvers carry full and partial performance descriptors with explicit, observable criteria.

Summary of Results

The finalized framework consists of one core examination, eight clusters, and presenting-problem pathways. Across clusters, learners add between 3 and 20 maneuvers beyond the core, averaging 8; musculoskeletal is organized by region, so a presentation draws only the group for the joint involved. Clusters are applied singly: each encounter is the core plus the one cluster indicated. In a preliminary comparison, one class taught under the framework was compared with three consecutive earlier classes taught head-to-toe, on a shared standardized-patient case at the same point in training. Restricted to the four indicated maneuvers scored with identical criteria in both years, complete performance rose from 72.3% to 88.2%, against an ordinary year-to-year variation of 8.9 percentage points.

Discussion and Conclusion

Separating universal maneuvers from context-specific additions aligns physical examination teaching with authentic practice while preserving foundational skills. Explicit performance descriptors establish a shared language for teaching, feedback, and assessment. The comparison is preliminary: one case, four maneuvers, non-randomised classes, and completeness rather than accuracy.